

BIG DATA ANALYTICS SOLUTIONS



ASSIGNMENT I

This assignment uses knowledge and skills from the previous few courses on Business Intelligence Solutions, Implementing Data Models and Reports, and Analyzing Data with Power BI.

This assignment will serve as a practical and hands on application with real world data. This is an individual assignment. You may not work with other students.

For this assignment, you will be provided with specific data by your instructor.

The assignment will be in two parts:

Assignment I Dashboard *and* a written report with supporting materials.

Assignment I Oral Presentation on Day 4

(note, your results might be used in Assignment II)

In this assignment, you will use Power BI to analyze and visualize data from the "Global Superstore" dataset, which represents the sales and profit data of a global retail company. You will create a dashboard and answer specific questions related to the data to gain valuable insights.

Task 1: Data Import and Transformation

Import the dataset into Power BI and perform the necessary data transformations, including:

- Handling missing values
- Renaming columns for clarity
- Formatting data types correctly

Task 2: Dashboard Creation

Create a Power BI dashboard that includes the following visualizations:

- A line chart showing the trend in total sales over time. Use the 'Order Date' column for the x-axis.
- A bar chart displaying the total profit for each product category.
- A map visualization showing sales by region (use 'Region' for the map).
- A scatter plot comparing discount rates (x-axis) to profit (y-axis).
- A table that lists the top 10 products by sales revenue.
- Show relationships between any 2 or 3 kinds of columns.

Task 3: Dashboard Questions

- Answer the following questions based on the data and the visualizations you created in Task 2:
 - What is the overall trend in sales over time? Are there any seasonal patterns?
 - Which product category is the most profitable? Which one is the least

- profitable?
- Which region generates the highest sales, and which one has the lowest sales?
- Is there any correlation between the country and profit? Explain.
- What are the top 10 best-selling products by sales revenue?

Your Oral Presentation will include the following:

Create a visually appealing and organized dashboard layout that includes the visualizations from Task 2 and the answers to the questions from Task 3. Ensure that the dashboard is easy to navigate and understand.

Your dashboard will be presented orally to the class and instructor at an assigned time for 20 minutes, plus 15 minutes of questions.

Consultation: The student will meet with the instructor once, on Day 2 or Day 3, at a mutually agreed upon time, for 30 minutes, to receive tips, suggestions, and consultation on the assignment progress.

Assignment I Presentation

- 20 minutes on Day 4, using any multimedia or presentation tools or the student's choosing
- 15 minutes of questions from the audience
 - **The presenter must prepare questions to engage the class, in case no questions are asked**
 - Students not conforming to time requirements will be penalized by up to 10% depending on the scale of non-conformance, either too long or too brief

Assignment I Dashboard Submission

- Dashboard is due immediately following the oral presentation
- Must contain solutions to Task 3.
- Must be not less than 500-750 words, 2 to 3 pages in length
 - Deductions made if it is too long or too short
- Submit electronically
 - Students not conforming to these requirements will be penalized by up to 10% depending on the scale of non-conformance.

Marking Scheme

- 10% Consultation I with the instructor
- 50% Dashboard submission
- 40% Oral Presentation of the Dashboard, showing grasp and professionalism.
 - Marks are based on
 - Presentation skills
 - Quality of the presentation
 - Ability to engage the class with asking and answering questions
 - Demonstrate capability to present the Dashboard with passion and knowledge to engage the audience
 - Communicate with ease and effective skills in order to reach the audience

- Make the presentation both interesting and fun

Spelling and Grammar: Proofread your submission for spelling and grammatical errors.